

APO-TERBINAFINE

Terbinafine Hydrochloride Tablets Equivalent to 250mg Terbinafine/Tablet Antifungal

Pharmacology

Terbinafine is an allylamine which has a broad spectrum of antifungal activity. At low concentrations, terbinafine is fungicidal against dermatophytes, molds and certain dimorphic fungi. The activity against yeasts is fungicidal or fungistatic, depending on the species.

Terbinafine interferes specifically with fungal sterol biosynthesis by inhibition of squalene epoxidase in the fungal cell membrane. The enzyme squalene epoxidase is not a cytochrome P450 enzyme. Its inhibition leads to a deficiency in ergosterol and to an intracellular accumulation of squalene, resulting in fungal cell death. When given orally, the drug concentrates rapidly in skin, hair and nails at levels associated with fungicidal activity.

Pharmacokinetics

A single oral dose of 250 mg terbinafine results in peak plasma concentrations of 0.97 mg/ml within 2 hours after administration. Seventy percent of the dose is absorbed by the gastrointestinal tract. The absorption half-life is 0.8 hours and the distribution half-life is 4.6 hours. The peak plasma concentration and bioavailability (AUC) roughly double when steady state is reached at 10-14 days. The bioavailability of terbinafine is moderately increased (20%) by food, but not sufficiently to require dosing adjustments.

Terbinafine binds strongly to plasma proteins (99%) and is lipophilic. Terbinafine is widely distributed in the body including adipose tissue. It rapidly diffuses through the dermis and concentrates in the lipophilic stratum corneum. Terbinafine is also secreted in sebum, thus achieving high concentrations in hair follicles, hair and in sebum-rich skin. There is evidence that terbinafine is distributed into the nail plate within the first few weeks of commencing therapy.

Terbinafine is excreted mainly in urine (80%) and in feces (20%). Following absorption terbinafine is metabolized rapidly and extensively by the liver. Terbinafine inhibits, but is not metabolized by CYP 2D6. Biotransformation is nearly complete and results in at least 15 identified metabolites all of which are excreted in the urine and lack antifungal activity. The plasma elimination half-life is 17 hours and the terminal half-life in stratum corneum, nail hair, dermis and epidermis ranges from 22 to 28 days. Higher plasma concentrations were noted in older adults and in hypertensives and lower concentrations were noted in smokers.

Indication

APO-TEBINAFINE (terbinafine hydrochloride) is indicated in the treatment of fungal infections of the skin and nails caused by dermatophytes such *Trichophyton* (e.g. *T. rubrum*, *T. mentagrophytes*, *T. verrucosum*, *T. violaceum*), *Microsporum canis* and *Epidermophyton floccosum*.

Oral APO-TERBINAFINE is indicated in the treatment of onychomycosis (fungal infection of the nail) caused by dermatophyte fungi.

APO-TERBINAFINE tablets may also be indicated in the treatment of tinea skin infections (tinea corporis, tinea cruris and tinea pedis).

Contraindication

APO-TERBINAFINE (terbinafine hydrochloride) tablets are contraindicated in patients with a hypersensitivity to terbinafine or any of the excipients.

Warnings

Rare cases of symptomatic hepatobiliary dysfunction including cholestatic hepatitis and very rarely liver failure have been reported.

If a patient presents with signs or symptoms of liver dysfunction such as unexplained persistent nausea, anorexia, tiredness,

abdominal pain, jaundice, pale stools or dark urine, treatment should immediately be discontinued and hepatic origin verified (See PRECAUTIONS, ADVERSE REACTIONS).

There have been isolated reports of anaphylactoid and serious skin reactions (e.g. Stevens Johnson Syndrome and Toxic Epidermal Necrolysis and Erythema Multiforme). If progressive skin rash occurs, treatment with terbinafine should be discontinued.

Isolated cases of blood dyscrasias have been reported in patients treated with terbinafine.

Precautions

There is very limited experience on the use of APO-TEBINAFINE in patients with liver disease or renal impairment. Plasma clearance appears to be reduced in some of these patients (see Pharmacokinetics).

Caution should be exercised in patients at higher risk of developing hepatic dysfunction (e.g. those receiving potentially hepatotoxic drugs; those with a history of alcoholism or suspicion of liver disorder).

Use in Pregnancy and Lactation: Though fetal toxicity and fertility studies have shown no adverse effects in animals, there is only very limited clinical experience with APO-TERBINAFINE (terbinafine hydrochloride) in pregnant women; therefore, unless the potential benefits outweigh any potential risks, APO-TERBINAFINE should not be used during pregnancy.

Terbinafine is excreted in breast milk; therefore mothers receiving oral treatment with APO-TERBINAFINE should not breast-feed.

Use in Elderly: When prescribing tablets for patients in this age group, the possibility of pre-existing impairment of liver or kidney function should be considered (see Pharmacokinetics).

Use in Pediatrics: There is limited experience with terbinafine in children.

Drug Interactions

The plasma clearance of terbinafine may be accelerated by drugs which induce metabolism (such as rifampicin) and may be inhibited by drugs which inhibit cytochrome P450 (such as cimetidine). Where co-administration of such agents is necessary, the dosage of Apo-Terbinafine may need to be adjusted accordingly.

Adverse Effects

Side effects are mild to moderate and transient. Gastrointestinal symptoms (feeling of fullness, loss of appetite, nausea, mild abdominal pain, diarrhoea) or skin reactions (rash, urticaria) being the most common ones. Isolated reports of serious, reversible skin reactions have been reported. If progressive morbilliform rash occurs, usage of APO-TERBINAFINE should be discontinued.

Infrequently, APO-TERBINAFINE tablet may cause taste disorder, including taste loss, which recover within several weeks after discontinuation of the drug.

There have been isolated cases of significant hepatobiliary dysfunction reported. Though a causal relationship to the drug has not been established, treatment with APO-TERBINAFINE tablet should be discontinued if hepatobiliary dysfunction develops. Isolated cases of neutropenia also have been reported.

Symptoms and Treatment of Overdose

A few cases of overdose with terbinafine of (up to 5g) have been reported, giving rise to headache, nausea, epigastric pain and dizziness. The recommended treatment of overdose consists of eliminating the drug primarily by the administration of activated charcoal and giving, symptomatic supportive therapy, if needed.

Route of Administration

Oral

Recommended Dose

APO-TERBINAFINE (terbinafine hydrochloride) Tablets: Adult dosage is 250 mg once daily.

The duration of treatment varies according to the indication and the severity of infection:

Indication	Duration of Treatment
Onychomycosis (of fingers and toes)*	6 weeks to 3 months
Skin infections**	
Tinea pedis (interdigital & plantar/moccasin type)	2 – 6 weeks
Tinea corporis, cruris	2 – 4 weeks

* In patients with fingernail infections or toenail infections other than the big toe, or in younger patients, treatment periods of less than 3 months may be adequate. In patients with infections of the big toenail, treatment for 3 months is usually sufficient, although some patients may require treatment for 6 months or longer. Poor nail outgrowth during the first weeks of treatment may enable identification of those patients in whom longer therapy is required.

In onychomycosis the optimal clinical effect is seen some months after mycological cure and cessation of treatment. This is related to the period required for outgrowth of healthy nail tissue.

** Complete resolution of the signs and symptoms may not occur until several weeks after mycological cure.

Availability of Dosage Forms

APO-TERBINAFINE 250 mg: Each round, white, biconvex, beveled-edged tablet, engraved 'APO' on one side, and scored and engraved 'TER' over '250' on the other contains terbinafine hydrochloride equivalent to 250 mg terbinafine. Available in PVC blisters of 30's.

Storage Conditions

Store below 30°C. Protect from light.

Shelf life

24 months from the date of manufacture.

Manufacturer

Apotex Inc., 150 Signet Drive, Toronto Ontario, Canada M9L 1T9.

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